Kainat Moin

AI & Machine Learning Engineer

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SUMMARY

AI & Machine Learning Engineer with hands-on experience in Python, Machine Learning, Deep Learning, Computer Vision, and AI application development. Skilled in building and evaluating intelligent systems, developing data-driven solutions, and applying AI techniques to real-world problems. Passionate about creating practical AI solutions and continuously advancing technical expertise.

EDUCATION

BS Artificial Intelligence
Dawood University of Engineering & Technology (DUET), ***Karachi | 2023 – Present***
Semester : *7* th
Current CGPA: 3.49

Intermediate (Pre-Engineering)
BAMM PECHS Government Girls College, ***Karachi | 2023 | Grade: A***

SKILL

- ***Programming Languages:*** Python, C++, C, SQL
- ***AI / Machine Learning:*** Machine Learning, Deep Learning, Computer Vision, CNN, Transfer Learning, Data Preprocessing, Feature Engineering, Model Evaluation
- ***Frameworks & Libraries:*** TensorFlow, Keras, Scikit-learn, OpenCV, MediaPipe, NumPy, Pandas, Matplotlib, Streamlit
- ***Web Technologies:*** HTML, CSS, WordPress
- ***Databases:*** MySQL, Oracle, Oracle APEX
- ***Tools & Platforms:*** Git, GitHub, Google Colab, Jupyter Notebook, VS Code, Semrush, Google Analytics, Search Console
- ***Core Strengths:*** Problem Solving, Analytical Thinking, Research, Team Collaboration, Communication, Project Management

EXPERIENCE

SEO & Content Writer

Ads-Expert | Feb 2025 – May 2025

- Created data-driven SEO content to improve search rankings and engagement.

Sales Executive

Groath Digital | April 2025 – June 2025

- Engaged clients via chat platforms to generate and convert leads.

CERTIFICATION

- *Google AI Professional Certificate – Google & Coursera | 2026*
- *Google AI Essentials Specialization – Google & Coursera | 2026*
- *Generative AI Application Developer (Top Performer) – NCEAC-HEC & ULEF, USA | 2026*
- *AI Got Talent – Certificate of Participation – FAST NUCES Karachi | 2026*
- *Proxion 2026 – Certificate of Participation – NED University | 2026*
- *Certificate of Participation in AI (ACIP)*
- *Web Development Professional Training Program*
- *Carbon Footprint Reduction Training Program*
- *Human Flag Formation Certificate of Participation*

PROJECT

PostPal — AI-Powered Fitness & Rehabilitation Assistant | Final Year Project (Ongoing)

Python | MediaPipe | OpenCV | TensorFlow Lite | LSTM | Computer Vision

- Developing an offline AI-powered fitness and rehabilitation application for exercise monitoring and guided movement.
- Using a device camera for real-time human pose estimation and exercise analysis.
- Providing voice-based guidance and real-time feedback to identify and correct exercise movements.
- Applying AI-based movement analysis for fitness and rehabilitation exercises while maintaining offline functionality.

Status: Ongoing — Final Year Project

NeuroFlow AI — NED Proxion 2026

- Developed an AI-powered career assistant for the NED Proxion 2026 competition.
- Built features for resume analysis, LinkedIn integration, interview preparation, and intelligent career guidance.
- Developed the application using React and JavaScript with AI/ML-oriented functionality.
- Collaborated on an AI-focused solution designed to support career development and job preparation.

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GitHub: https://github.com/kainatmoin/NeuroFlow-Proxion2026.git
***Live Demo*:** https://neuro-flow-psi.vercel.app/

Emotion Classifier – AI Got Talent (DevDay '26, FAST NUCES Karachi) | Python, Scikit-Learn, Streamlit

- Built a multi-label emotion classifier to detect 9 emotions from tweets in a 2-hour hackathon.
- Applied TF-IDF feature extraction and an ensemble of machine learning models.
- Achieved F1 Score: 89.3%, Precision: 88.6%, and Recall: 90.1%.
- Deployed the model as a live web application.

GitHub: https://github.com/kainatmoin/Tweet-Emotion-Classifier.git
***Live Demo*:** https://aigot-talent-urpxliqh6zzqzwg5gqtbpa.streamlit.app/

Human Pose Estimation & Activity Classification Pipeline | MediaPipe, OpenCV, Python

- Developed a pose estimation system for detecting human body landmarks from video.
- Computed joint angles (knee, hip, shoulder) for activity analysis.
- Classified activities (Squat, Standing, Transition) using a rule-based model.
- Achieved 94.3% accuracy in activity classification.

Link: https://github.com/kainatmoin/Human-Pose-Estimation-Activity-Classification.git

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Plant Disease Classification using Deep Learning | TensorFlow, Keras, OpenCV

- Developed an image-based plant disease classification system for 38 classes using CNN and Transfer Learning.
- Implemented CNN from Scratch and MobileNetV2 (Frozen & Fine-Tuned) models for comparison.
- Applied image preprocessing and augmentation using OpenCV and TensorFlow.
- Achieved 92.06% accuracy with Fine-Tuned MobileNetV2 and built a prediction pipeline in Google Colab.

Link: https://github.com/kainatmoin/Plant-Disease-Classification-Using-Deep-Learning.git

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Waste Classification using Image Clustering

- Built waste classification system using MobileNetV2 (CNN) for 9 classes
- Applied PCA for dimensionality reduction and K-Means for clustering
- Evaluated using Silhouette Score & Davies–Bouldin Index
- Developed prediction pipeline in Google Colab

Link:https://github.com/kainatmoin/Smart-Waste-Classification.git

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